

Design and performance analysis of an annular metal hydride reactor for large-scale hydrogen storage applications

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ABSTRACT

The present study is focused on developing a modular design of a metal hydride reactor for large-scale hydrogen storage applications. In this perspective, an annular metal hydride reactor that is cooled/heated on both inner and outer surfaces is designed and analyzed. A numerical model is developed to study the absorption and desorption characteristics of the designed annular metal hydride reactor. Three configurations of the annular metal hydride reactor that could offer a weight ratio (i.e., mass of metal hydride alloy to mass of the reactor) of 2 are investigated. The first two configurations are of annular metal hydride reactor with only change in the heat transfer fluid's flow direction. The third configuration is the same as the second one, with extra radial fins inside the metal hydride bed. The main objective of studying the three configurations is to visualize the enhancement achieved in the reaction rate and outlet temperature. The results showed that the peak outlet temperature is enhanced by 3.6 °C with the second configuration. At the same time, the third configuration improved the absorption and desorption rates by a factor of 2.07 and 1.92, respectively, while the radial fins occupying only 4.6% of reactor volume. Further, a sensitivity analysis is conducted on the third configuration to analyze the influence of operating parameters on absorption and desorption times. Furthermore, the performance of the third configuration is compared with several designs present in the literature.

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1. Introduction

In recent years, metal hydrides (MHs) are considered a promising technology for many engineering applications, viz hydrogen storage, hydrogen compression, heating and cooling, and thermal energy storage [1]. The MH-based hydrogen storage system requirements are high energy storage density, fast charging and discharging rates, and cyclic stability [2]. The heat transfer rate to/from the MH bed is a controlling factor that determines the reaction rate of the MH bed rather than the mass transfer rate [3]. The MH reactor's performance is mainly affected by the internal heat transfer constraint imposed by the low thermal conductivity of the MH bed and the geometrical parameters (for example, thickness of the bed) of the reactor. Hence, designing an efficient MH reactor should aim to attain faster charging and discharging rates and achieve a higher weight ratio (i.e., the ratio of the mass of MH powder to the empty reactor mass). In this regard, several

researchers proposed performance enhancement techniques that include novel heat exchanger configurations and adding thermal conductivity enhancers to MH bed [4,5].

An experimental study on spiral heat exchanger revealed that the absorption time is reduced by a factor of 5 compared to the reactor cooled by natural convection [6]. In a numerical study of an annulus disc reactor embedded with cooling tubes, the absorption time was reduced by 60% when six cooling tubes were used instead of four cooling tubes [7]. A finned tube hydrogen storage device filled with LaNi₅ could reach 1.2 wt% capacity in 490 s at a supply pressure of 15 bar and a cooling fluid temperature of 15 °C [8]. Another research was focused on investigating the effect of heat transfer coefficient and bed thickness on the hydrogenation performance of rectangular MH tanks with embedded cooling tubes (ECTs) [9]. The results indicated an optimum value for the MH bed thickness was 10.39 mm, and a heat transfer coefficient over 2000 W/m².K did not enhance the hydrogenation rate by more than 5%. Andreasen et al. [10] experimentally investigated the discharge behavior of an MH container equipped with internal (radial) and external (axial) fins. The MH container's response was observed when immersed in a thermostatic water bath and exposed to air.

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The use of external fins has positive results when exposed to air rather than immersed in water. They suggested that the effectiveness of a particular MH container should be evaluated for fixed external convective conditions. On the other hand, several researchers [11–14] studied the behavior of an MH tank equipped with phase change materials. The performance of these systems is mainly affected by latent heat and the thermal conductivity of PCM. However, the reaction completion time achieved with PCM jacketed MH reactor was more than 5000 s, which is not competitive with the other existing reactor designs.

Further, the details of large-scale metal hydride reactors reported in the literature are presented in Table 1. It is observed that most of the large-scale MH hydrogen tanks are of shell and tube type [15–18]. Also, the shell and tube-type reactors equipped with internal fins reduce the number of cooling tubes to achieve equivalent performance. A few studies [19,20] were focused on the reactors equipped with internal fins that were cooled from the outer surface. However, only internal fins without any cooling tube could not deliver better performance than the ECT reactor. From this, it is understood that the finned tubes enhance the reaction kinetics effectively instead of multiple ECTs.

Several MH reactor geometries such as cylindrical tubes, shell and tube, internal annular rings, coiled tubes, rectangular tanks, and microchannel reactors were studied in the literature [4,5]. However, those reactors are comprised of a simple design with slower reaction rates or a complex design with reasonably faster reaction rates. Further, the reported designs of large-scale MH reactors [15,16,18–20] are complex, involve fabrication difficulties, and offer a lower weight ratio. Hence, a simple reactor design with reasonably higher reaction rates for large-scale hydrogen storage applications is essential. In this perspective, an annular MH reactor cooled/heated on both the inner and outer surfaces that could offer a simple design, higher weight ratio, and better performance is

proposed. Also, previous studies have focused only on the absorption and desorption rates but not on the outlet temperature of heat transfer fluid (HTF). Therefore, with the help of the annular MH reactor, we propose a way to enhance the outlet temperature of HTF (or difference between the inlet and outlet temperatures of the HTF) without hampering the reaction rate for the first time. The higher the outlet temperature of the HTF, the higher is the exergy output.

In the present study, a 2D mathematical model is developed in COMSOL Multiphysics to visualize the absorption and desorption behavior of annular MH reactors. The model solves for velocity and temperature field of HTF, which helps to analyze the outlet temperature of the HTF. The metal hydride material is selected to be LaNi_5 as it is an extensively studied material, and thus, enables an easier comparison of the present results with the literature. Three configurations of the annular MH reactor differing by flow directions of the HTF and the absence/presence of radial fins are studied. Further, a sensitivity analysis of the annular MH reactor is performed to study the effect of operating parameters on absorption and desorption characteristics. Furthermore, the present study results are compared with the literature, and a large-scale MH reactor layout assembly of several annular MH reactors studied in this work has been outlined.

2. Physical system and mathematical formulation

2.1. Physical system

An annular MH reactor is designed for a working pressure of 80 bar. The annular MH reactor provides better cooling/heating rates than solid cylindrical reactors. The annular MH reactor is cooled/heated on both the internal and the external surfaces using an inner cooling tube and an outer cooling jacket. The 2D sectional

Table 1
Large scale metal hydride reactors studied in the literature.

Author	Design	Alloy used	Observations/Remarks	Reference
Johnson et al. (2011)	Tube bundle reactor (total four modules)	86 kg of sodium alanate–graphite mixture	- Each module consisted of 12 tubes arranged in a staggered array. Each tube was filled with 1.79 kg of MH. - The system was designed to store 3 kg of hydrogen. - The system was operated between 120 and 220 °C. - The desorption rate of 2 g/s was achieved.	[15]
Gkanas et al. (2016)	Shell and tube	LaNi_5	- Compared the performance of ECT reactor with and without fins. - The reactor with 12 longitudinal finned tubes showed the equivalent performance to the reactor with 60 ECT.	[16]
Kumar et al. (2019)	Shell and tube	40 kg of $\text{LaNi}_{4.7}\text{Al}_{0.3}$	- The reactor consisted of 99 cooling tubes. - Under 15 bar supply pressure, 425.7 g of H_2 was absorbed in 1017 s.	[17]
Karmakar et al. (2020)	Shell and tube	10 kg of LaNi_5	- At 80 °C, the alloy desorbed 395.812 g of H_2 in 3720 s. - The cooling system consisted of four cooling tubes and an outer cooling jacket. - The MH bed reached 1.12 wt% in 1980 s under a supply pressure of 15 bar. - At 80 °C, complete desorption happened in 2700 s.	[18]
Lototskiy et al. (2020)	MH cassette encased in the lead (each cassette comprising of several MH containers embedded with perforated copper fins)	AB_2 type	- A hydrogen storage tank for heavy-duty fuel cell utility vehicles was developed. - The tank could supply the hydrogen at 120 NL/min to operate an 11-kW fuel cell stack. - The refueling time of the MH tank was about 15–20 min at the supply conditions: $T = 15\text{--}20$ °C and $P_{\text{H}_2} = 100\text{--}150$ bar.	[19]
Afzal et al. (2021)	Reactor with hexagonal honeycomb inserts	47.5 kg of $\text{La}_{0.9}\text{Ce}_{0.1}\text{Ni}_5$	- The reactor equipped with hexagonal honeycomb fins was cooled from the outer surface. - Under 25 bar supply pressure, the reactor consumed 5400 s to reach a reacted fraction of 0.9. - At 50 °C of heating fluid temperature, the desorption time was 10000 s.	[20]

view of the annular MH reactor is shown in Fig. 1. An axial disc filter is fitted near the reactor's left wall for passing H_2 gas in/out of the reactor. The present study selects commercially available ASTM A321 standard tubes of SS316 material to design the reactor. A 2.5-inch nominal size with an outer diameter and thickness of 73.03 mm and 2.11 mm, respectively, is chosen for the reactor. The inner and outer diameter of the central cooling tube is 18.04 mm and 21.34 mm, respectively. The thickness of the outer cooling jacket is considered as 4 mm. $LaNi_5$ is selected as MH alloy for the present analysis and its thermo-physical properties are shown in Table 2. Here, three different configurations of annular MH reactors are studied and compared. The details of the three configurations studied are shown in Fig. 1.

2.1.1. Configuration 1

Configuration 1 is simple annular geometry with cooling/heating provided on inner and outer surfaces. It consists of two inlets and one outlet for the HTF. The HTF enters at inlet temperature through the inlet ports provided in the inner cooling tube and outer cooling jacket and exits from the common outlet, as shown in Fig. 1 (a). The total mass flow rate of the HTF is distributed as per the cross-sectional area of the inner cooling tube and outer cooling

Table 2

Thermo-physical properties of $LaNi_5$ and hydrogen [21–23].

Property	$LaNi_5$	Hydrogen
Molecular weight (g/mol)	434	2.01
Specific heat (J/kg. K)	419	14283
Thermal conductivity (W/m. K)	1.32	0.127
Reaction Enthalpy, ΔH (J/mol)	–29879	–
Reaction Constant, C (1/s)	59.187 (absorption) 9.57 (desorption)	–
Activation Energy, E (J/mol)	21170 (absorption) 16445 (desorption)	–
Density (kg/m ³)	8400	–
Maximum H_2 storage capacity (wt.% _{max})	1.379	–
Porosity	0.5	–
Permeability (m ²)	10^{-8}	–
Universal gas constant (J/mol. K)	8.314	–
van't Hoff constants: A, B [K]	13.44, 3780	–
Constants:		
Slope factors, (φ_s) (φ_o)	0.038	–
	0	–
Hysteresis factor (β)	0.137	–

jacket. A weight ratio defined in Eq. (1) signifies the thermal mass of the whole reactor. The weight ratio of the MH reactor depends on

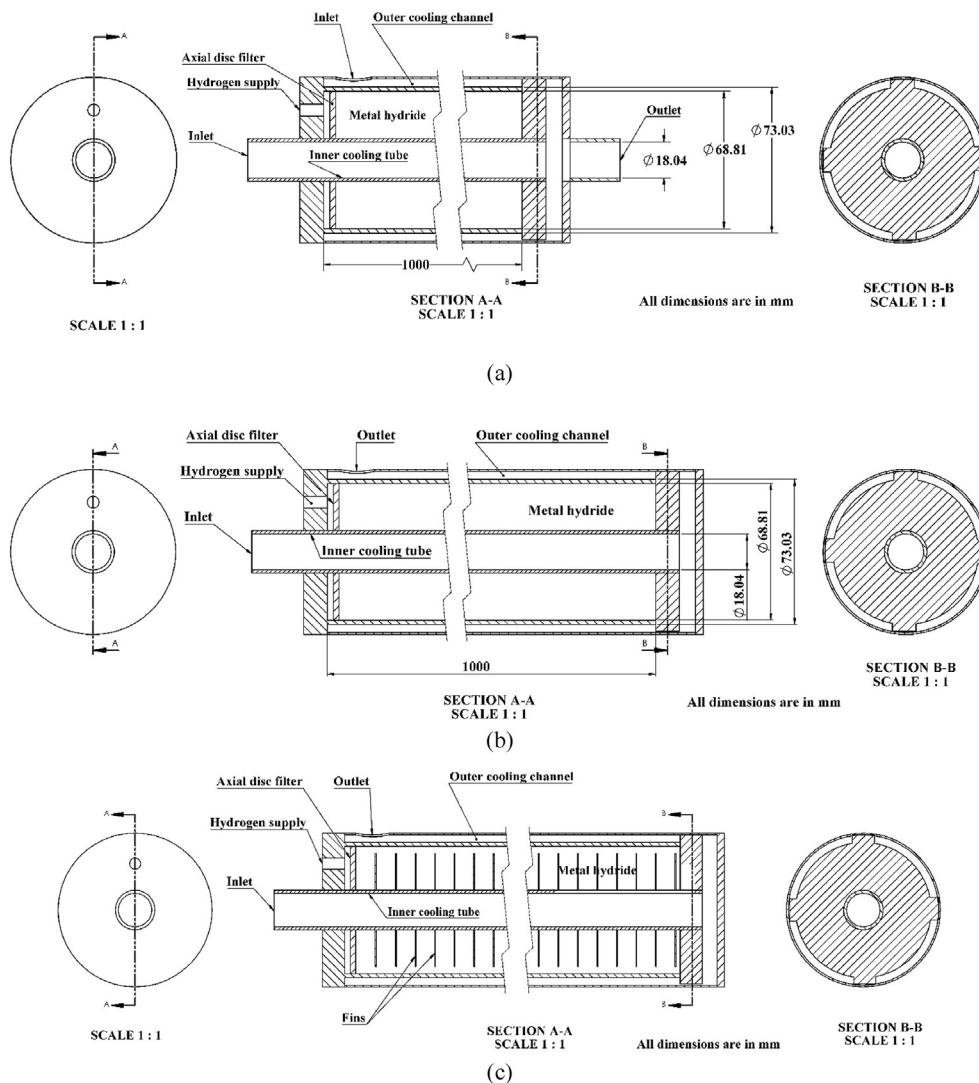


Fig. 1. Annular MH reactors studied (a) configuration 1 (b) configuration 2 (c) configuration 3.

the working pressure of the reactor and the heat transfer enhancement provided. The weight ratio offered by configuration 1 is 2.14 for the working pressure of 80 bar. In the present study, stainless steel material is considered for the outer cooling jacket. However, the weight ratio can be further increased by making the outer cooling jacket with plastic or aluminium alloy.

$$\text{Weight ratio} = \frac{\text{Mass of metal hydride alloy}}{\text{Mass of the empty reactor}} \quad (1)$$

Mass of the empty reactor includes the mass of the reactor, cooling tube, outer cooling jacket, fins, and endplates.

2.1.2. Configuration 2

A double pass heat exchanger concept is introduced to achieve a higher peak in the HTF outlet temperature for the given mass flow rate, as shown in Fig. 1 (b). The reactor consists of one inlet and one outlet for the HTF. The HTF enters through the inlet of the inner cooling tube and exits from the outlet of the outer cooling jacket. This design is simple and reduces the number of HTF ports. However, there is a slight increase in the pressure drop in HTF flow due to double pass. The weight ratio for configuration 2 is calculated as 2.14. The purpose of studying configuration 2 is to evaluate the improvement in reaction times and the rise in the outlet temperature of HTF compared to configuration 1.

2.1.3. Configuration 3

Configuration 3 is similar to configuration 2 with additional radial fins inside the MH reactor. Since internal heat transfer is a limiting factor for the MH reactor, radial fins are added to enhance the heat transfer rate to/from the MH reactor from/to the HTF. Aluminum is selected as a fin material due to its lightweight and good thermal conductivity. A total of 95 fins (thickness 0.5 mm, pitch 10.4 mm) are placed inside the reactor. The schematic of configuration 3 is shown in Fig. 1 (c). The radial fins occupied 4.6% of the reactor's total volume, due to which there is a 6% decrease in the weight ratio (i.e., 2.01). The key objective of considering configuration 3 is to analyze the heat transfer rate enhancement achieved in the MH due to the addition of fins.

The mass of metal hydride present in configuration 1, configuration 2, and configuration 3 is 14.1, 14.1 and 13.5 kg, respectively. There is a slight reduction in the mass of metal hydride present in configuration 3 due to the addition of fins.

2.2. Mathematical formulation

In this study, a 2D mathematical model is formulated for the absorption and desorption processes in the annular MH reactor as the problem's solution is axisymmetric (Fig. 2). Mass, momentum, and energy conservation laws are applied in the porous material. Hydrogen is charged/discharged from the top of the reactor during the absorption/desorption, as shown in Fig. 2. The annular MH reactor is cooled/heated during the absorption/desorption processes by supplying the HTF. The flow behavior of the HTF is simulated instead of considering the heat transfer coefficient on the heat exchanger's outer walls.

2.2.1. Assumptions

The following assumptions are associated with the governing equations for the absorption and desorption processes [24].

1. Hydrogen behaves as an ideal gas
2. Local thermal equilibrium is valid between MH and hydrogen gas
3. Metal hydride material is homogeneous and isotropic

4. Porosity and permeability are uniform
5. The thermophysical properties of MH are independent of temperature, concentration, and pressure.
6. The expansion and contraction of metal hydride during hydriding and dehydriding processes, respectively, are neglected.

2.2.2. Governing equations

Hydride mass balance:

The following equation specifies the mass conservation of the solid metal hydride.

$$(1 - \varepsilon) \frac{\partial \rho_s}{\partial t} = \dot{m} \quad (2)$$

Hydrogen gas mass balance:

Variation of hydrogen gas density in the MH is due to hydrogen flow and simultaneous absorption/desorption in metal hydride. Mass balance equation for hydrogen gas is given by Eq. (3).

$$\varepsilon \frac{\partial \rho_g}{\partial t} + \frac{1}{r} \frac{\partial}{\partial r} (r \rho_g u_r) + \frac{\partial}{\partial z} (\rho_g u_z) = - \dot{m} \quad (3)$$

The reaction rate during absorption and desorption are given by Eqs. (4) and (5), respectively.

$$\dot{m} = C_a \exp\left(-\frac{E_a}{R_u T}\right) \ln\left(\frac{P_g}{P_{eq}}\right) (\rho_{sat} - \rho_s) \quad (\text{Absorption}) \quad (4)$$

$$\dot{m} = C_d \exp\left(-\frac{E_d}{R_u T}\right) \left(\frac{P_g - P_{eq}}{P_{eq}}\right) (\rho_s - \rho_o) \quad (\text{Desorption}) \quad (5)$$

Momentum equation

Darcy's law describes the velocity of the hydrogen in porous metal hydride.

$$u_r = \frac{-K}{\mu_g} \frac{\partial P_g}{\partial r} \quad (6)$$

$$u_z = \frac{-K}{\mu_g} \frac{\partial P_g}{\partial z}$$

where the pressure, P_g satisfies the ideal gas equation

$$P_g = \rho_g R_g T \quad (7)$$

Energy equation

$$(\rho C_p)_e \frac{\partial T}{\partial t} + \rho_g C_{pg} u_r \frac{\partial T}{\partial r} + \rho_g C_{pg} u_z \frac{\partial T}{\partial z} = \frac{1}{r} \frac{\partial}{\partial r} \left(r k_e \frac{\partial T}{\partial r} \right) + \frac{\partial}{\partial z} \left(k_e \frac{\partial T}{\partial z} \right) - (1 - \varepsilon) \dot{m} \left(\frac{\Delta H}{M_{H_2}} \right) \quad (8)$$

Equilibrium pressure:

In reality, metal hydrides exhibit a pressure hysteresis during hydrogen absorption and desorption due to the development of mechanical strain in the metal lattice with the insertion of hydrogen atoms into interstitial sites. Metal hydrides also exhibit sloping behavior in the plateau region of pressure-composition-isotherm due to inhomogeneities in composition. Nishizaki et al. (1983) proposed an equation (Eq. (9)) for equilibrium pressure by curve fitting the experimental P-C-T data that considers the hysteresis and slope effects into account [25].

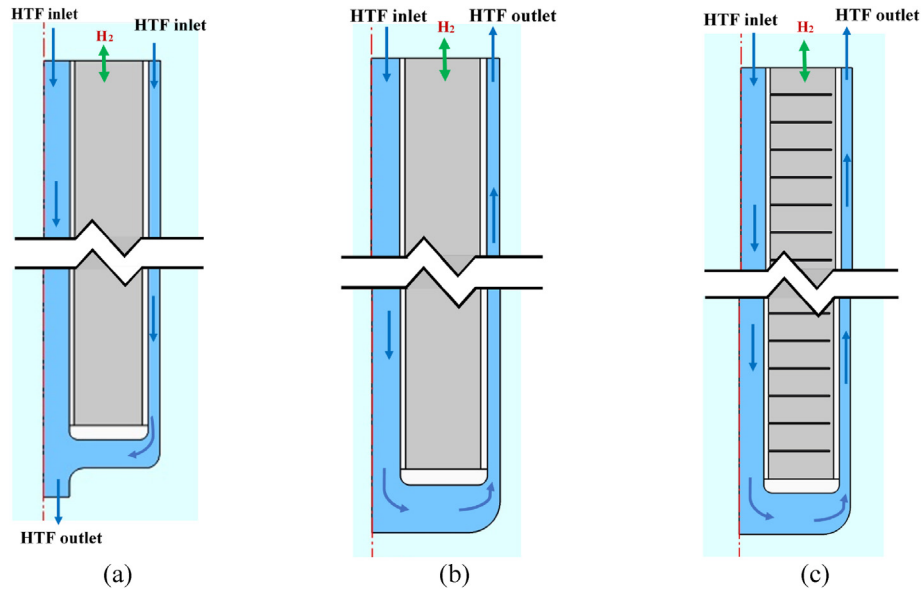


Fig. 2. Two-dimensional axisymmetric computational domain (not to scale) (a) configuration 1, (b) configuration 2 and (c) configuration 3.

$$P_{eq} = \exp \left\{ \left[A - \frac{B}{T} \right] + (\phi_s \pm \phi_0) \tan \left[\pi \left(\frac{c}{c_{sat}} - \frac{1}{2} \right) \right] \pm \frac{\beta}{2} \right\} \times 10^5 \quad (9)$$

Here the sign '+' is for absorption and '-' is for desorption.

The relations between concentration, hydride density, and wt.% are given as follows [26].

$$\text{Concentration, } c = \frac{2(\rho_s - \rho_o)M_{MH}}{\rho_o M_{H_2} N_{MH}} \quad (10)$$

$$\begin{aligned} \frac{\partial u_r}{\partial t} + u_r \frac{\partial u_r}{\partial r} + u_z \frac{\partial u_r}{\partial z} &= -\frac{1}{\rho_f} \frac{\partial p}{\partial r} + \frac{\mu_f}{\rho_f} \left\{ -\frac{u_r}{r^2} + \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u_r}{\partial r} \right) + \frac{\partial^2 u_r}{\partial z^2} \right\} \\ \frac{\partial u_z}{\partial t} + u_r \frac{\partial u_z}{\partial r} + u_z \frac{\partial u_z}{\partial z} &= -\frac{1}{\rho_f} \frac{\partial p}{\partial z} + \frac{\mu_f}{\rho_f} \left\{ \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u_z}{\partial r} \right) + \frac{\partial^2 u_z}{\partial z^2} \right\} \end{aligned} \quad (13)$$

Energy equation:

$$\rho_f C_{pf} \left(\frac{\partial T}{\partial t} + u_r \frac{\partial T}{\partial r} - u_z \frac{\partial T}{\partial z} \right) = \frac{1}{r} \frac{\partial}{\partial r} \left(r k_f \frac{\partial T}{\partial r} \right) + \frac{\partial}{\partial z} \left(k_f \frac{\partial T}{\partial z} \right) \quad (14)$$

Governing equations for reactor tube:

The conduction heat transfer phenomena inside the solid

$$\text{Weight percentage, wt.\%} = \frac{\text{mass of hydrogen absorbed/desorbed (g)}}{\text{mass of alloy used (g)}} = \frac{100(\rho_s - \rho_o)}{\rho_o} \quad (11)$$

Governing equations for HTF:

Water is used as a heat transfer fluid. The governing equations for incompressible fluid flow are as follows.

Continuity equation:

$$\frac{1}{r} \frac{\partial (r u_r)}{\partial r} + \frac{\partial u_z}{\partial z} = 0 \quad (12)$$

Momentum equation:

domain (reactor tube, HTF tube, and fins) is modeled using the following energy equation.

$$\rho_{sd} C_{psd} \frac{\partial T}{\partial t} = \frac{1}{r} \frac{\partial}{\partial r} \left(r k_{sd} \frac{\partial T}{\partial r} \right) + \frac{\partial}{\partial z} \left(k_{sd} \frac{\partial T}{\partial z} \right) \quad (15)$$

2.2.3. Initial and boundary conditions

Initial conditions:

The initial conditions used in the present work during absorption and desorption are presented in Table 3.

Boundary conditions:

The boundary conditions used in the model are presented in Table 4. The inlet pressure is ramped/reduced from the initial equilibrium pressure to the supply/discharge pressure using Eq. (16) to avoid convergence issues.

Table 3

Initial conditions.

Metal hydride	Heat transfer fluid
Absorption: $T(r, z, 0) = T_{o,a}$ $P(r, z, 0) = P_{eqi,a}$ $\rho_s(r, z, 0) = \rho_{si,a}$ $c(r, z, 0) = c_{i,a}$	Desorption: $T(r, z, 0) = T_{o,d}$ $P(r, z, 0) = P_{eqi,d}$ $\rho_s(r, z, 0) = \rho_{si,d}$ $c(r, z, 0) = c_{i,d}$
	$T(r, z, 0) = T_{o,a/o,d}$ $P(r, z, 0) = P_o$ $u_r(r, z, 0) = u_z(r, z, 0) = 0$

Table 4

Boundary conditions.

Type	Boundary condition
Adiabatic condition on the outer walls	$\frac{\partial T}{\partial n} = 0$
No-slip and non-permeability on the walls of HTF tube	$u_r = u_z = 0$
Hydrogen inlet (supply pressure for absorption and discharge pressure for desorption) and No flow condition	At the inlet of the reactor, $p = P_{\text{supply/discharge}}$
Inlet temperature and velocity for HTF	On walls of MH reactor, $\frac{\partial p}{\partial n} = 0$ or $\vec{n} \cdot \rho \vec{V} = 0$
Outlet and outflow condition for HTF	$u_z = U_{\text{inlet}}$ $T = T_{\text{inlet}}$ $p = P_o$
Heat flux continuity	$\frac{\partial T}{\partial n} = 0$ On the interface between heat exchanger tube and MH reactor: $-k_{MH} \nabla T _{MH} = -k_{sd} \nabla T _{sd}$ On the interface between HTF and heat exchanger tube: $-k_{sd} \nabla T _{sd} = -k_f \nabla T _f$

$$P_{\text{inlet}}(t) = P_{eqi} + D(t) (P_{\text{supply/discharge}} - P_{eqi}) \quad (16)$$

Where $D(t)$ is a step function defined at $t = 0.05$ s.

2.2.4. Numerical modeling

The numerical solution for Eqs. (2–15) subjected to the above-mentioned initial and boundary conditions have been obtained using the finite element-based simulation tool, COMSOL Multiphysics 5.3. Non-isothermal flow Multiphysics coupling is used between fluid flow and heat transfer phenomenon. All the simulations are performed on a Dell Precision T7610 workstation, equipped with two Intel Xeon E5-2650 v2 processors and 64 GB RAM. The resulting linear equations are solved using the segregated approach with a direct linear solver, PARDISO. The relative and absolute tolerance values of 5×10^{-4} and 2.5×10^{-5} are used, respectively, in the simulation.

2.3. Model validation

The present numerical results are validated with Muthukumar et al. [27] (MH reactor with outer cooling jacket) and Singh et al. [28] (MH reactor with finned tube), due to lack of experimental data on the annular MH reactor that is cooled/heated both on the inner and outer surfaces in the literature. The results obtained and the corresponding operating conditions used for the validation are shown in Fig. 3. It is observed that the predicted numerical results are in good agreement with the experimental data.

3. Results and discussion

This section presents the comparison between the performance of three reactor configurations during the absorption and desorption processes. The supply pressure is set to 15 bar during absorption since a commercial electrolyzer typically supplies hydrogen at 15 bar [9]. Simultaneously, the inlet temperature and flow rate of the HTF are maintained at 25 °C and 2.5 lpm,

respectively. During desorption, the discharge pressure of the hydrogen is set to 1 bar gauge pressure. The inlet temperature and the flow rate of the HTF are set to 50 °C and 2.5 lpm, respectively. For all the simulations, initial concentration of hydrogen of the MH bed is considered as 0.049 and 0.939 during absorption and desorption, respectively. The results of three configurations are compared in terms of the average temperature of the MH bed, weight percentage, and outlet temperature of HTF. Further, the sensitivity analysis results are presented for the finned tube reactor (configuration 3). Furthermore, the performance of the finned tube reactor is compared with the literature.

3.1. Comparison between reactor configurations during absorption

Absorption is an exothermic reaction during which the MH bed releases heat by absorbing the hydrogen. The average bed temperature, amount of hydrogen absorbed in terms of wt.%, and the outlet temperature of HTF are compared for all three configurations during absorption, as shown in Fig. 4. The MH bed absorbed 0.25 wt % in 5s for all three configurations due to the rapid rate of absorption (Fig. 4a). During the initial stages of the reaction, the driving potential (difference between the supply pressure and the equilibrium pressure of the bed) for mass transfer is very high, which caused higher rates of absorption. The rate of exothermic heat generation is immensely higher than the rate of heat removal by the HTF during the initial stages of the reaction. Due to this, the bed temperature increases and reaches a peak value. The peak temperature attained by the three reactor configurations is approximately 69 °C. The increase in the bed temperature causes a decrease in the driving potential, reducing the absorption rate. On the other hand, the heat transfer rate between the MH bed and the HTF increases due to increased bed temperature. After that, the temperature of the MH bed drops and tends to reach the inlet temperature of the HTF. The average bed temperature of configuration 3 reached inlet HTF temperature in 1105 s, whereas the same for configuration 2 is 1665 s (Fig. 4a). From Fig. 4a, it is observed that the time required to absorb 90% of the maximum storage

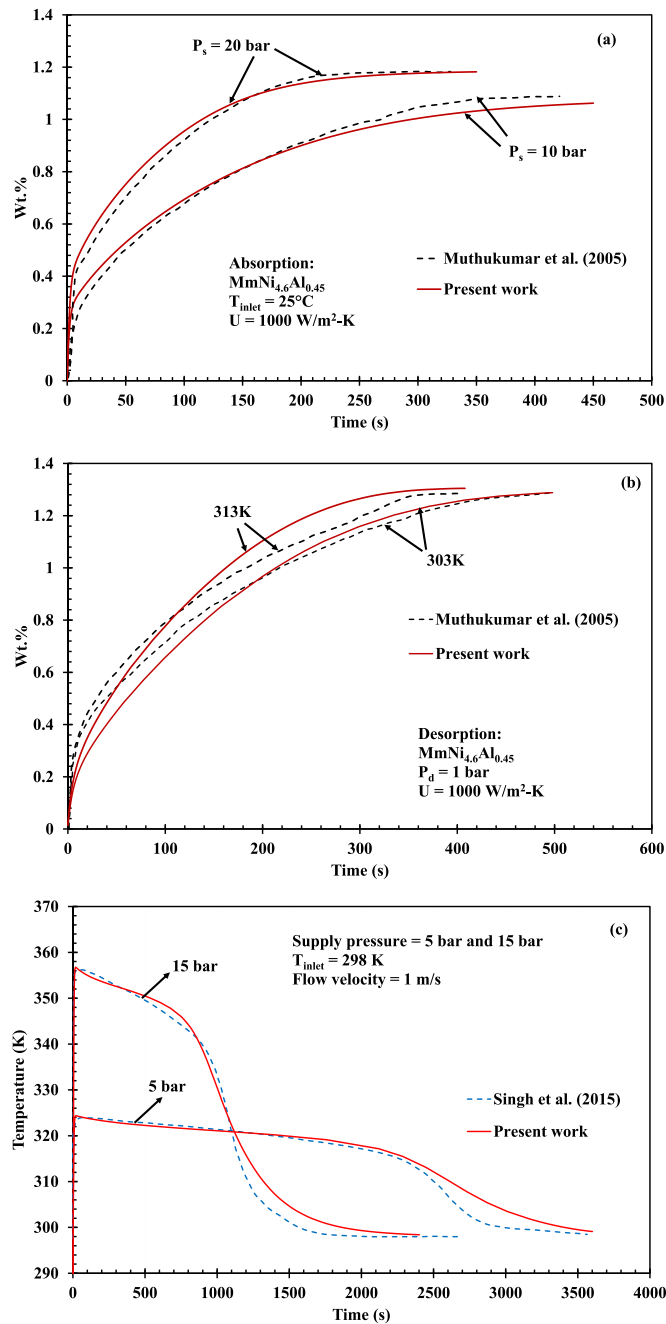


Fig. 3. Validation of the present numerical model with experimental work (a) absorption, (b) desorption of Muthukumar et al. (2005) and (c) temperature profile of Singh et al. (2015).

capacity is 646, 587, and 283 s, respectively, for configuration 1, configuration 2, and configuration 3. The absorption time of configuration 2 is decreased by 9.1% compared to configuration 1. This is because the mass flow rate of the HTF at any particular cross-section of the inner cooling tube and the outer cooling jacket is higher for configuration 2 compared to configuration 1. Whereas configuration 3 showed a decrement of 51.8% in absorption time compared to configuration 2 due to its high heat transfer area provided by the addition of radial fins. Configuration 3 is 2.07 times faster than configuration 2 to reach 90% of the maximum storage capacity, while fins occupy only 4.6% of the reactor volume.

The outlet temperature of the HTF for three reactor

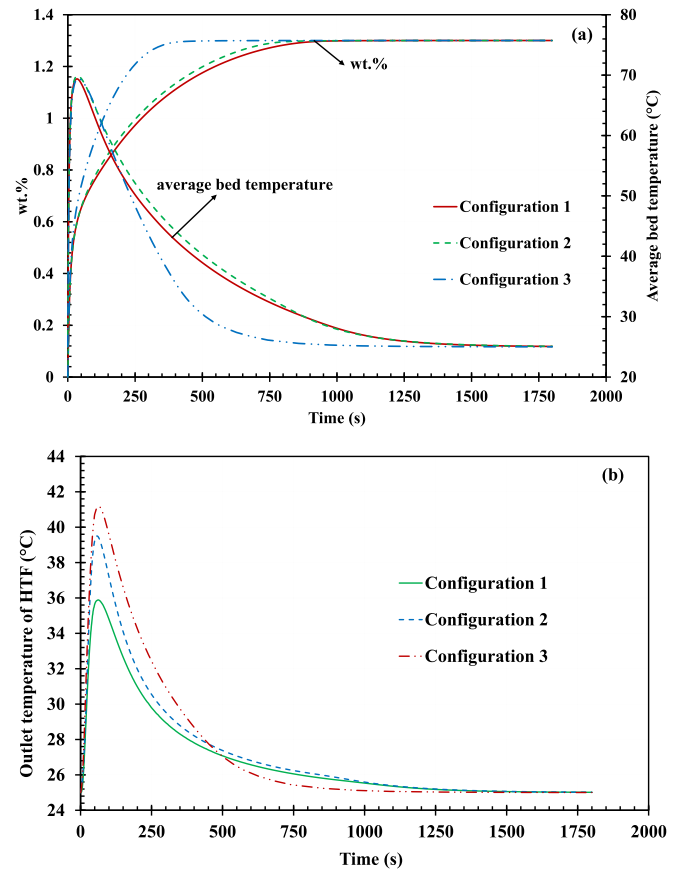


Fig. 4. Comparison between three reactor configurations during absorption (a) average bed temperature and weight percentage and (b) outlet temperature of HTF.

configurations during absorption is compared in Fig. 4b. It is observed that the outlet temperature of the HTF is improved with configuration 2 than configuration 1. The peak outlet temperature attained by configuration 1 and configuration 2 is 35.89 and 39.5°C , respectively. The main reason for enhancement in the HTF outlet temperature is the double pass effect of the HTF. However, it could be noticed that the total amount of energy extracted from both configurations 1 and 2 is the same. Configuration 3 also enhanced the peak in the HTF outlet temperature to 41.17°C . The average temperature difference between the inlet and outlet temperatures of HTF obtained during absorption are 4.4, 5.7 and 10.6 , respectively, for configuration 1, configuration 2 and configuration 3. The absorption results suggest that configuration 3 is recommended for achieving faster reaction rates and higher outlet temperatures of HTF.

3.2. Comparison between the reactor configurations during desorption

Desorption is an endothermic process during which the MH bed releases hydrogen by absorbing the heat from HTF. The average bed temperature, amount of hydrogen desorbed (wt.%) and the outlet temperature of HTF are compared during desorption, as shown in Fig. 5. Initially, the temperature of the MH bed is at 50°C . The inlet temperature and flow rate of HTF are 50°C and 2.5 lpm, respectively. The MH bed desorbed 0.25 wt% in 8 s for all three configurations due to the rapid rate of desorption (Fig. 5a). During the initial stages, there is a sudden drop in the temperature of the MH bed due to the rapid endothermic reaction. The rapidity of the

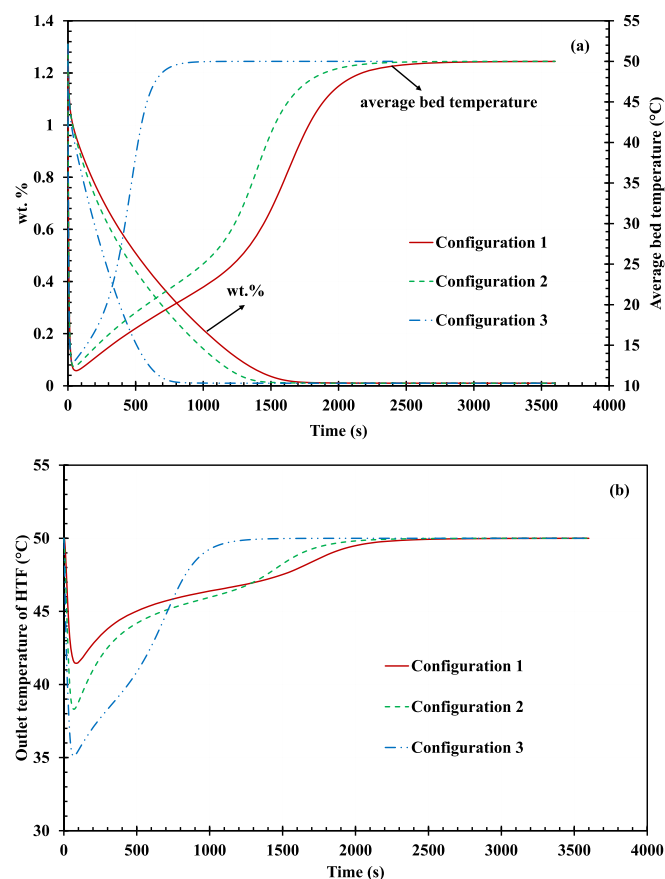


Fig. 5. Comparison between the reactor configurations during desorption (a) average bed temperature and weight percentage and (b) outlet temperature of HTF.

endothermic reaction is due to higher driving potential. The rate of endothermic heat absorption is much higher than the heat transfer rate from the HTF to the MH bed during the early stages of the reaction. Due to this, the bed temperature decreases and reaches the lowest value. The average temperature of the MH bed dropped to the lowest value of 11.86, 12.36, 12.84 °C for the case of configuration 1, configuration 2, and configuration 3, respectively, during the early stages (within 70 s) of the desorption (Fig. 5a). The decrease in the bed temperature causes a reduction in the driving potential, resulting in a reduction in the desorption rate. On the other hand, the heat transfer rate between the MH bed and the HTF increases due to the decrease in the bed temperature. After that, the temperature of the MH bed rises and tends to reach the inlet temperature of the HTF. The average bed temperature of configuration 3 reached inlet HTF temperature in 1292 s, whereas the same for configuration 2 is 3340 s (Fig. 5a). From Fig. 5a, it is observed that the time required to desorb 90% of the maximum storage capacity is 1172, 1010, and 525 s, respectively, for configurations 1, configurations 2, and configurations 3. Configuration 2 showed a 16% improvement in the desorption rate compared to configuration 1. The reason for the improvement is explained in section 3.1. On the other hand, configuration 3 showed 92.4% improvement in the desorption rate compared to configuration 2 due to its high heat transfer area provided by the addition of radial fins. Configuration 3 is 1.92 times faster than configuration 2 to desorb 90% of the stored hydrogen.

The outlet temperature of the HTF for three reactor configurations during desorption is compared in Fig. 5b. The lowest value of the outlet temperature attained by configuration 1, configuration 2

and configuration 3 is 41.45, 38.3 and 35.1 °C, respectively. Further, the average temperature difference between the inlet and outlet temperatures of HTF obtained during desorption is 4.99, 6.38 and 11.6 °C, respectively, for configuration 1, configuration 2, and configuration 3. Similar to the absorption process, it is also observed that the enhancement in the outlet temperature of HTF is achieved during desorption with configuration 2 and configuration 3 compared to configuration 1 for the same mass flow rate of the HTF.

From the average bed temperature profiles (Figs. 4a and 5a), it is observed that radial fins have less influence during the early stages of the absorption and desorption processes. During the later stages, the average bed temperature of configuration 3 reached the HTF inlet temperature at a faster rate due to the increased heat transfer rate than configuration 1 and configuration 2. With the change in the flow configuration, there is an enhancement in the reaction rate and outlet temperature of HTF (Figs. 4b and 5b). On the other hand, the pressure drop in the HTF flow is increased from 44.7 Pa (configuration 1) to 54 Pa (configuration 2 and 3). Configuration 2 enhanced the peak in the outlet temperature of the HTF is by 3.6 °C while the pressure drop is increased by only 9.3 Pa. In this regard, the present design's pressure drop is compared with the ECT reactor studied by Kumar et al. [17]. The pressure drop inside an ECT (inner diameter = 4.57 mm; length = 1000 mm) is varied from 43 Pa (for the total flow rate of 10 lpm) to 129 Pa (for the total flow rate of 30 lpm). From this analysis, it is understood that the pressure drop of configuration 3 is in proximity with the ECT reactor.

3.3. Sensitivity analysis

The effect of the supply/discharge pressure, inlet temperature and mass flow rate of the HTF on time required to reach 90% of the storage/discharge capacity during absorption/desorption (referred to as absorption/desorption time) is presented in this section. The effect of the mass flow rate of the HTF and supply pressure on absorption time is shown in Fig. 6a, when the inlet temperature of the HTF is fixed at 25 °C. The absorption time is reduced by increasing both the supply pressure and the mass flow rate of the HTF. It is observed that the supply pressure has a significant effect on lowering the absorption time than the mass flow rate of the HTF. The percentage decrease in the absorption time is in the range of 55.5%–57.98% when the supply pressure is increased from 10 to 20 bar for the considered range of mass flow rate of the HTF, whereas the same is 15.75%–22.36% when the mass flow rate of the HTF is increased from 0.0167 to 0.0333 kg/s for the considered range of supply pressure. Further, increasing the mass flow rate of the HTF is effective at lower supply pressures than higher supply pressures, as seen in Fig. 6a. The percentage reduction in absorption time when the mass flow rate of the HTF is increased from 0.0167 kg/s to 0.05 kg/s is 32% at the supply pressure of 10 bar, whereas the same is 23% at the supply pressure of 30 bar. The effect of the inlet temperature of the HTF and the supply pressure on absorption time is shown in Fig. 6b, when the mass flow rate of the HTF is fixed at 0.042 kg/s. The absorption time is reduced by increasing the supply pressure and decreasing the inlet temperature of the HTF. The absorption time is reduced by 34% when the inlet temperature of HTF is reduced from 35 to 25 °C for the supply pressure of 10 bar, whereas the same is 28.9% for the supply pressure of 30 bar.

The effect of the heating fluid temperature and mass flow rate of the HTF on desorption time is shown in Fig. 7 when the discharge pressure of the hydrogen is fixed at 1 bar (gauge). The desorption time is reduced by increasing both the heating fluid temperature and the mass flow rate of the HTF. It is observed that the heating fluid temperature has a significant effect on lowering the

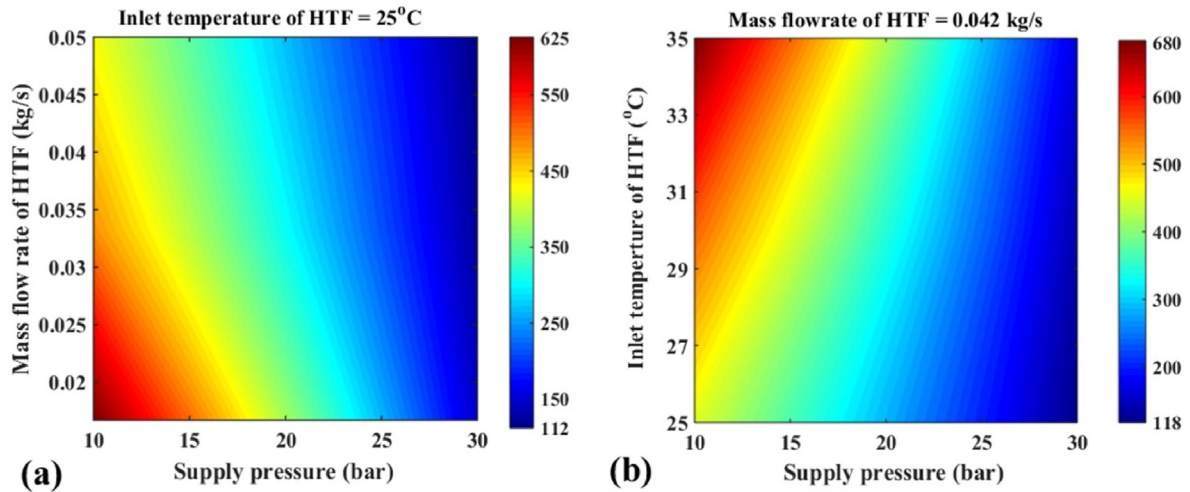


Fig. 6. Effect of supply pressure, inlet temperature and mass flow rate of HTF on absorption time.

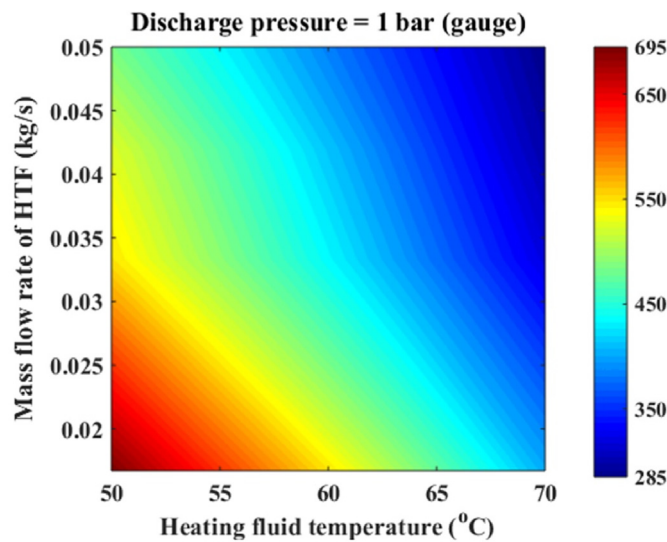


Fig. 7. Effect of heating temperature and mass flow rate of HTF on desorption time.

desorption time than the mass flow rate of the HTF. The percentage decrease in the desorption time is in the range of 41%–42% when the heating fluid temperature is increased from 50 °C to 70 °C for the considered range of mass flow rate of the HTF. The percentage reduction in desorption time is 30.2% when the mass flow rate of

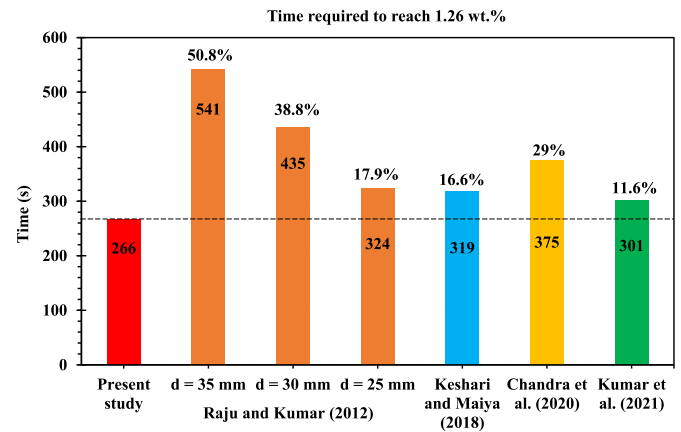


Fig. 8. Comparison of configuration 3 with literature during absorption.

the HTF is increased from 0.0167 kg/s to 0.05 kg/s at the heating fluid temperature of 50 °C, whereas the same is 29.2% at the heating fluid temperature of 70 °C.

Further, the sensitivity contours explicitly explain the relative influence of the operating parameters on the magnitude of the reaction completion times. These are effective in selecting suitable operating parameters for the efficient operation of MH reactors.

Table 5

Metal hydride reactors filled with LaNi₅ that are considered for the comparison.

Author	Type of reactor	Operating conditions/remarks	Ref.
Present study	Configuration 3: Finned tube (single tube with 95 radial fins) reactor with outer cooling jacket	$P_s = 15$ bar ; $T_{inlet} = 25^\circ\text{C}$; $\dot{m}_{HTF} = 3$ lpm	–
Raju and Kumar (2012)	Tube bundle reactor: One single tube is modeled with following dimensions: Inner diameter = 25–35 mm Thickness = 2 mm; Length = 1000 mm	Modeled under similar conditions of the present study with LaNi ₅ as [29] storage material	
Keshari and Maiya (2018)	Pin fin tube heat exchanger: 16 cooling tubes	$P_s = 15$ bar ; $T_{inlet} = 25^\circ\text{C}$; $\dot{m}_{HTF} = 6.75$ lpm (the result is directly taken)	[30]
Chandra et al. (2020)	Multi-tube reactor with conical fins (6 tubes and 19 fins)	$P_s = 15$ bar ; $T_{inlet} = 20^\circ\text{C}$; $\dot{m}_{HTF} = 5.9$ lpm (the result is directly taken)	[31]
Kumar et al. (2021)	Embedded cooling tube reactor: 55 cooling tubes	Modeled under similar conditions of the present study with LaNi ₅ as [32] storage material	

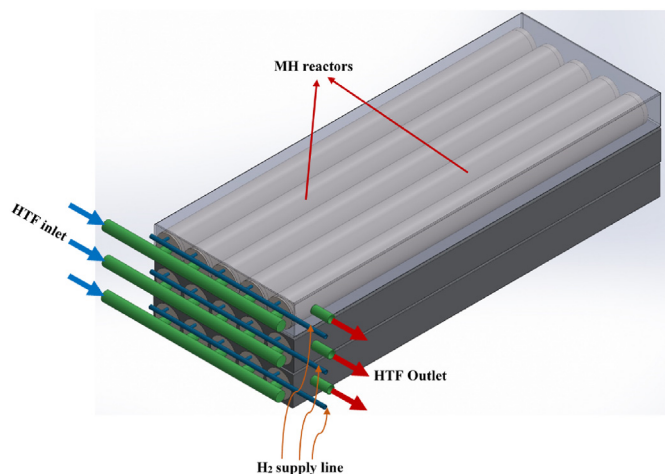


Fig. 9. Proposed modular design for large scale hydrogen storage applications.

3.4. Comparison of the performance of configuration 3 with the literature

The absorption performance of configuration 3 is compared with relevant studies from the literature. The details of the reactors considered for the comparison and their operating conditions are shown in Table 5. Raju and Kumar (2012) and Kumar et al. (2021) studied different MH material than LaNi_5 . Hence, the present comparison study adopted the reactor design by them (Raju and Kumar 2012 and Kumar et al., 2021) and analyzed the performance with LaNi_5 as a storage material using the numerical model developed in this study. The reactors are modeled under similar working conditions and the mass flow rate of the HTF is selected based on the mass of the alloy filled inside the reactor. The time to reach 1.26 wt% during absorption is compared, and the percentage reduction in the absorption time obtained by configuration 3 over the other reactors is shown in Fig. 8.

Configuration 3 (finned tube reactor) showed improved performance over the multi-tubular reactors. The heat transfer rate is enhanced because of the radial fins and the annular bed geometry of configuration 3. From the comparative study, it is understood that providing heat transfer enhancement inside the MH bed is more effective than using multiple ECTs. On the other hand, for the tube bundle reactor, tube diameter of less than 25 mm is required to obtain the comparative performance with the present design, which increases the total number of tubes inside the tube bundle reactor. As the number of tubes increases inside the ECT reactor/tube bundle reactor, the pressure drop in the HTF flow and fabrication complexity also increase. Hence, the authors believe that the proposed annular MH reactor equipped with radial fins is a competitive design for large-scale hydrogen storage applications.

4. Proposed modular design of large-scale MH reactor

A modular design for a large-scale MH-based hydrogen storage system is proposed in this section. The main objectives of the modular design are to reduce the total weight of the MH reactor and easy fabrication without compromising the thermal performance. The present modular design concept considers several small reactors connected in parallel rather than a single large reactor, as shown in Fig. 9. The annular MH reactor equipped with radial fins is parallelly arranged inside each module. Since several annular metal hydride reactors are connected in parallel, there exists a minimum loss of alloy due to any accidental leakage of HTF

into the reactor. In a single large-scale hydrogen storage unit, all the MH material present in the system needs to be heated or cooled irrespective of the hydrogen demand, which results in large parasitic heat loss. With the modular design, each module can be operated independently as per the demand, resulting in less parasitic heat loss and a better response time.

5. Conclusions

A numerical model is developed for investigating the absorption and desorption characteristics of the annular MH reactor filled with LaNi_5 . Three reactor configurations are compared to analyze the effect of fluid flow direction and adding radial fins on the absorption and desorption processes. Further, a sensitivity analysis is carried out on the finned tube reactor to study the impact of operating parameters on absorption and desorption times. Furthermore, the performance of the finned tube reactor is compared with the literature. Based on the results of the present numerical study, the following key observations are made:

- The absorption time of 283 s is obtained with the finned tube reactor under the supply pressure of 15 bar and cooling fluid temperature of 25 °C. During desorption, the finned tube reactor achieved a desorption time of 525 s under the discharge pressure of 1 bar and heating fluid temperature of 50 °C.
- The addition of fins enhanced the absorption and desorption rates by a factor of 2.07 and 1.92, respectively.
- Changing the HTF flow direction, enhanced the peak outlet temperature by 3.6 °C for configuration 2.
- From sensitivity analysis, it is observed that the supply pressure and desorption temperature are the most influencing parameters during the absorption and desorption processes, respectively.
- The absorption performance of the configuration 3 is compared with the literature. Configuration 3 (finned tube reactor) showed improved performance over the ECT reactor and tube bundle reactor. In addition, the heat transfer rate is enhanced much better with radial fins inside the annular bed than with the multi-tube reactor.

Finally, a modular design of annular metal hydride reactors is proposed for large-scale hydrogen storage applications.

CRediT authorship contribution statement

J. Sunku Prasad: Conceptualization, Methodology, Investigation, Software, Writing – original draft. **P. Muthukumar:** Validation, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Nomenclature

A	material constant
B	material constant (K)

c	concentration
C	reaction constant (s^{-1})
C_p	specific heat ($J\ kg^{-1}\ K^{-1}$)
E	activation energy ($J\ mol^{-1}$)
k	thermal conductivity ($W\ m^{-1}\ K^{-1}$)
K	permeability (m^2)
$\dot{m}$	hydrogen mass transferred per unit time and unit volume ($kg\ m^{-3}\ s^{-1}$)
M	molecular weight ($kg\ mol^{-1}$)
N	Number of atoms
R_g	specific gas constant for hydrogen ($J\ mol^{-1}\ K^{-1}$)
R_u	universal gas constant ($J\ kg^{-1}\ K^{-1}$)
t	time (s)
T	temperature (K)
u, v	velocities ($m\ s^{-1}$)
$wt.\%$	weight percentage
r, z	coordinate axes

Greek letters

β	hysteresis factor
ΔH	reaction enthalpy ($J\ mol^{-1}$)
ϵ	porosity
μ	dynamic viscosity (Pa. s)
ϕ_s	slope factor
ϕ_o	slope constant
ρ	density ($kg\ m^{-3}$)

Subscripts

a	absorption
d	desorption
e	effective
eq	equilibrium
f	heat transfer fluid
g	gas
H_2	hydrogen gas
i	initial
o	empty
s	solid/hydride
sat	saturation
sd	solid domain

Acronyms

MH	Metal Hydride
HTF	Heat Transfer Fluid

References

- [1] P. Muthukumar, A. Kumar, N.N. Raju, K. Malleswararao, M.M. Rahman, A critical review on design aspects and developmental status of metal hydride based thermal machines, *Int. J. Hydrogen Energy* (2018) 1–27, <https://doi.org/10.1016/j.ijhydene.2018.07.157>.
- [2] G. Han, Y.K. Kwon, J.B. Kim, S. Lee, J. Bae, E.A. Cho, B.J. Lee, S. Cho, J. Park, Development of a high-energy-density portable/mobile hydrogen energy storage system incorporating an electrolyzer, a metal hydride and a fuel cell, *Appl. Energy* 259 (2020), 114175, <https://doi.org/10.1016/j.apenergy.2019.114175>.
- [3] F. Yang, X. Meng, J. Deng, Y. Wang, Z. Zhang, Identifying heat and mass transfer characteristics of metal hydride reactor during adsorption-Parameter analysis and numerical study, *Int. J. Hydrogen Energy* 33 (2008) 1014–1022, <https://doi.org/10.1016/j.ijhydene.2007.11.007>.
- [4] A. Mazzucco, M. Dornheim, M. Sloth, T.R. Jensen, J.O. Jensen, M. Rokni, Bed geometries, fueling strategies and optimization of heat exchanger designs in metal hydride storage systems for automotive applications: a review, *Int. J. Hydrogen Energy* 39 (2014) 17054–17074, <https://doi.org/10.1016/j.ijhydene.2014.08.047>.
- [5] S. Shafiee, M.H. McCay, Different reactor and heat exchanger configurations for metal hydride hydrogen storage systems - a review, *Int. J. Hydrogen Energy* 41 (2016) 9462–9470, <https://doi.org/10.1016/j.ijhydene.2016.03.133>.
- [6] S. Mellouli, F. Askri, H. Dhaoui, A. Jemni, S. Ben Nasrallah, A novel design of a heat exchanger for a metal-hydrogen reactor, *Int. J. Hydrogen Energy* 32 (2007) 3501–3507, <https://doi.org/10.1016/j.ijhydene.2007.02.039>.
- [7] A. Boukhari, R. Bessaïh, Numerical heat and mass transfer investigation of hydrogen absorption in an annulus-disc reactor, *Int. J. Hydrogen Energy* 40 (2015) 13708–13717, <https://doi.org/10.1016/j.ijhydene.2015.05.123>.
- [8] A. Singh, M.P. Maiya, S. Srinivasa Murthy, Experiments on solid state hydrogen storage device with a finned tube heat exchanger, *Int. J. Hydrogen Energy* 42 (2017) 15226–15235, <https://doi.org/10.1016/j.ijhydene.2017.05.002>.
- [9] E.I. Gkanas, M. Khzouz, G. Panagakos, T. Statheros, G. Mihalakakou, G.I. Siasos, G. Skodras, S.S. Makridis, Hydrogenation behavior in rectangular metal hydride tanks under effective heat management processes for green building applications, *Energy* 142 (2018) 518–530, <https://doi.org/10.1016/j.energy.2017.10.040>.
- [10] G. Andreasen, M. Melnichuk, S. Ramos, H.L. Corso, A. Visintin, W.E. Triaca, H.A. Peretti, Hydrogen desorption from a hydride container under different heat exchange conditions, *Int. J. Hydrogen Energy* 38 (2013) 13352–13359, <https://doi.org/10.1016/j.ijhydene.2013.07.115>.
- [11] H. El Mghari, J. Huot, J. Xiao, Analysis of hydrogen storage performance of metal hydride reactor with phase change materials, *Int. J. Hydrogen Energy* 44 (2019) 28893–28908, <https://doi.org/10.1016/j.ijhydene.2019.09.090>.
- [12] T. Alqahtani, S. Mellouli, A. Bamasag, F. Askri, P.E. Phelan, Thermal performance analysis of a metal hydride reactor encircled by a phase change material sandwich bed, *Int. J. Hydrogen Energy* 45 (2020) 23076–23092, <https://doi.org/10.1016/j.ijhydene.2020.06.126>.
- [13] S.N. Nyamsi, I. Tolj, M. Lototsky, Metal hydride beds-phase change materials: dual mode thermal energy storage for medium-high temperature industrial waste heat recovery, *Energies* 12 (2019), <https://doi.org/10.3390/en12203949>.
- [14] S.S. Ardahaie, M.J. Hosseini, M. Eisapour, A.H. Eisapour, A.A. Ranjbar, A novel porous metal hydride tank for hydrogen energy storage and consumption assisted by PCM jackets and spiral tubes, *J. Clean. Prod.* 311 (2021) 127674, <https://doi.org/10.1016/j.jclepro.2021.127674>.
- [15] T.A. Johnson, S.W. Jorgensen, D.E. Dedrick, Performance of a full-scale hydrogen-storage tank based on complex hydrides, *Faraday Discuss* 151 (2011) 327–352, <https://doi.org/10.1039/c0fd00017e>.
- [16] E.I. Gkanas, D.M. Grant, M. Khzouz, A.D. Stuart, K. Manickam, G.S. Walker, Efficient hydrogen storage in up-scale metal hydride tanks as possible metal hydride compression agents equipped with aluminium extended surfaces, *Int. J. Hydrogen Energy* 41 (2016) 10795–10810, <https://doi.org/10.1016/j.ijhydene.2016.04.035>.
- [17] A. Kumar, N.N. Raju, P. Muthukumar, P.V. Selvan, Experimental studies on industrial scale metal hydride based hydrogen storage system with embedded cooling tubes, *Int. J. Hydrogen Energy* 44 (2019) 13549–13560, <https://doi.org/10.1016/j.ijhydene.2019.03.180>.
- [18] A. Karmakar, A. Mallik, N. Gupta, S. Pratibha, Studies on 10kg alloy mass metal hydride based reactor for hydrogen storage, *Int. J. Hydrogen Energy* (2020), <https://doi.org/10.1016/j.ijhydene.2020.11.091>.
- [19] M. Lototsky, I. Tolj, Y. Klochko, M.W. Davids, D. Swanepoel, V. Linkov, Metal hydride hydrogen storage tank for fuel cell utility vehicles, *Int. J. Hydrogen Energy* 45 (2020) 7958–7967, <https://doi.org/10.1016/j.ijhydene.2019.04.124>.
- [20] M. Afzal, N. Gupta, A. Mallik, K.S. Vishnula, S. Pratibha, Experimental analysis of a metal hydride hydrogen storage system with hexagonal honeycomb-based heat transfer enhancements-part B, *Int. J. Hydrogen Energy* 46 (2021) 13131–13141, <https://doi.org/10.1016/j.ijhydene.2020.11.275>.
- [21] R. Busqué, R. Torres, J. Grau, V. Roda, A. Husar, Effect of metal hydride properties in hydrogen absorption through 2D-axisymmetric modeling and experimental testing in storage canisters, *Int. J. Hydrogen Energy* 42 (2017) 19114–19125, <https://doi.org/10.1016/j.ijhydene.2017.06.125>.
- [22] K. Malleswararao, A. N. S. Srinivasa Murthy, P. Dutta, Performance prediction of a coupled metal hydride based thermal energy storage system, *Int. J. Hydrogen Energy* (2020), <https://doi.org/10.1016/j.ijhydene.2020.03.251>.
- [23] K. Jiao, X. Li, Y. Yin, Y. Zhou, S. Yu, Q. Du, Effects of various operating conditions on the hydrogen absorption processes in a metal hydride tank, *Appl. Energy* 94 (2012) 257–269, <https://doi.org/10.1016/j.apenergy.2012.01.033>.
- [24] A.A. Rabienataj Darzi, H. Hassanzadeh Afrouzi, A. Moshfegh, M. Farhadi, Absorption and desorption of hydrogen in long metal hydride tank equipped with phase change material jacket, *Int. J. Hydrogen Energy* 41 (2016) 9595–9610, <https://doi.org/10.1016/j.ijhydene.2016.04.051>.
- [25] T. Nishizaki, K. Miyamoto, K. Yoshida, Coefficients of performance of hydride heat pumps, *J. Less Common. Met.* 89 (1983) 559–566, [https://doi.org/10.1016/0022-5088\(83\)90372-7](https://doi.org/10.1016/0022-5088(83)90372-7).
- [26] R. Busqué, R. Torres, J. Grau, V. Roda, A. Husar, Mathematical modeling, numerical simulation and experimental comparison of the desorption process in a metal hydride hydrogen storage system, *Int. J. Hydrogen Energy* 43 (2018) 16929–16940, <https://doi.org/10.1016/j.ijhydene.2017.12.172>.
- [27] P. Muthukumar, M.P. Maiya, S.S. Murthy, Experiments on a metal hydride-based hydrogen storage device, *Int. J. Hydrogen Energy* 30 (2005) 1569–1581, <https://doi.org/10.1016/j.ijhydene.2004.12.007>.
- [28] A. Singh, M.P. Maiya, S.S. Murthy, Effects of heat exchanger design on the performance of a solid state hydrogen storage device, *Int. J. Hydrogen Energy* 40 (2015) 9733–9746, <https://doi.org/10.1016/j.ijhydene.2015.06.015>.
- [29] M. Raju, S. Kumar, Optimization of heat exchanger designs in metal hydride based hydrogen storage systems, *Int. J. Hydrogen Energy* 37 (2012) 2767–2778, <https://doi.org/10.1016/j.ijhydene.2011.06.120>.
- [30] V. Keshari, M.P. Maiya, Design and investigation of hydriding alloy based hydrogen storage reactor integrated with a pin fin tube heat exchanger, *Int. J.*

- Hydrogen Energy 43 (2018) 7081–7095, <https://doi.org/10.1016/j.ijhydene.2018.02.100>.
- [31] S. Chandra, P. Sharma, P. Muthukumar, S.S.V. Tatiparti, Modeling and numerical simulation of a 5 kg LaNi5-based hydrogen storage reactor with internal conical fins, Int. J. Hydrogen Energy 45 (2020) 8794–8809, <https://doi.org/10.1016/j.ijhydene.2020.01.115>.
- [32] A. Kumar, N.N. Raju, P. Muthukumar, Parametric studies on MmNi4.7Fe0.3 based reactor with embedded cooling tubes for hydrogen storage and cooling application, J. Energy Storage 35 (2021) 102317, <https://doi.org/10.1016/j.est.2021.102317>.